

ROSLAVL, Russian Federation

WMO: 268820

Lat: 53.933N

Lon: 32.833E

Elev: 224

StdP: 98.66

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.8	-18.7	-24.6	0.4	-21.4	-21.4	0.6	-18.2	10.6	-0.7	9.4	-1.1	2.3	110	0.565

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	28.7	19.9	26.9	19.1	25.2	18.3	21.0	26.8	20.1	25.3	19.2	23.9	2.7	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.0	14.2	23.7	18.1	13.4	22.5	17.3	12.7	21.6	61.5	26.9	58.4	25.3	55.4	24.1	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	7.6	6.7	-25.2	30.9	2.8	2.4	-27.2	32.7	-28.9	34.1	-30.5	35.5	-32.6	37.2		
(5)			-25.4	22.4	2.8	0.9	-27.4	23.0	-29.0	23.6	-30.6	24.1	-32.7	24.8	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.2	-6.3	-5.7	-0.7	7.1	13.0	16.5	19.0	17.2	11.8	6.0	0.3	-4.3	(6)	
	DBStd	10.10	6.07	6.18	4.60	4.66	4.65	3.51	3.16	3.47	3.65	4.08	4.93	5.91	(7)	
	HDD10.0	2307	506	441	333	112	23	1	0	0	22	136	290	444	(8)	
	HDD18.3	4532	764	674	591	337	174	77	30	63	197	382	540	703	(9)	
	CDD10.0	927	0	0	1	25	116	195	278	224	77	11	0	0	(10)	
	CDD18.3	111	0	0	0	0	9	21	50	29	2	0	0	0	(11)	
	CDH23.3	862	0	0	0	3	72	148	378	246	15	0	0	0	(12)	
	CDH26.7	194	0	0	0	1	10	20	92	71	1	0	0	0	(13)	
(14) Wind	WSAvg	3.2	3.7	3.7	3.7	3.2	3.0	2.9	2.4	2.6	2.7	3.2	3.5	3.7	(14)	
(15) Precipitation	PrecAvg	604	32	24	30	36	47	87	84	61	58	54	48	39	(15)	
	PrecMax	793	76	59	73	88	126	149	178	182	190	122	97	82	(16)	
	PrecMin	410	2	2	1	4	3	14	15	11	5	0	0	9	(17)	
	PrecStd	102	17	13	17	21	24	37	40	43	39	30	21	17	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.7	4.9	14.2	22.5	28.1	28.6	31.5	32.8	25.3	19.5	11.9	8.1	(19)	
		MCWB	3.9	2.9	7.8	12.4	19.1	19.1	21.1	20.6	17.1	12.6	10.3	7.5	(20)	
	2%	DB	2.9	3.1	10.0	19.7	25.5	26.9	29.1	28.2	22.1	15.8	9.2	5.6	(21)	
		MCWB	2.1	2.0	5.2	11.4	17.1	19.1	20.2	20.2	15.6	12.9	8.2	4.9	(22)	
	5%	DB	1.6	2.0	7.5	17.4	23.1	25.0	27.1	25.9	19.8	13.5	7.7	3.4	(23)	
		MCWB	1.0	1.0	4.0	10.4	16.1	18.0	19.6	18.6	14.7	11.1	6.8	2.8	(24)	
	10%	DB	0.6	1.2	5.2	15.0	21.0	23.2	25.3	23.8	17.8	11.8	6.5	2.0	(25)	
		MCWB	0.1	0.5	2.6	9.2	14.3	17.4	18.9	17.6	13.6	9.9	5.7	1.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	3.7	8.2	13.6	19.7	20.9	22.5	22.0	18.7	15.1	10.8	7.6	(27)	
		MCDB	4.3	4.5	13.0	19.8	26.2	26.4	29.0	28.1	24.2	17.5	11.5	7.9	(28)	
	2%	WB	2.2	2.2	6.0	12.1	18.1	19.9	21.3	20.7	16.7	13.0	8.4	4.9	(29)	
		MCDB	2.7	2.8	9.0	17.7	23.9	24.9	27.2	26.6	21.0	15.1	8.9	5.4	(30)	
	5%	WB	1.1	1.3	4.3	11.1	16.8	19.0	20.4	19.7	15.3	11.5	7.1	2.9	(31)	
		MCDB	1.5	1.8	6.9	16.1	21.8	23.8	25.5	25.1	18.3	13.2	7.6	3.3	(32)	
	10%	WB	0.2	0.7	2.8	9.9	15.5	17.9	19.4	18.3	14.1	10.2	5.7	1.5	(33)	
		MCDB	0.6	1.1	4.8	14.4	19.9	22.2	24.0	22.4	16.9	11.5	6.4	1.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	5.4	6.5	8.8	9.6	9.0	9.4	9.6	8.7	6.4	4.1	4.0	(35)	
		MCDBR	3.5	4.4	9.6	12.1	12.4	11.2	11.9	12.6	11.9	8.9	5.0	3.9	(36)	
	5% WB	MCWBR	3.5	3.8	5.8	5.9	5.8	5.1	4.8	5.4	6.2	5.9	4.2	3.6	(37)	
		MCWBR	3.3	4.0	8.5	10.6	11.3	10.2	10.7	11.5	10.0	8.1	4.3	3.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.289	0.324	0.349	0.415	0.393	0.377	0.408	0.421	0.376	0.350	0.328	0.292	(40)	
		taud	2.132	2.072	2.125	2.144	2.303	2.366	2.299	2.252	2.348	2.380	2.255	2.152	(41)	
	Ebn at Noon		643	740	816	808	853	872	833	795	788	724	599	557	(42)	
		Edh at Noon	67	100	118	134	121	115	121	120	96	75	61	54	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.70	1.61	3.04	4.01	5.39	5.70	5.49	4.54	3.02	1.46	0.55	0.41	(44)	
	RadStd		0.15	0.26	0.41	0.41	0.37	0.51	0.48	0.50	0.30	0.30	0.12	0.11	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (9 neighbors)		+0.64	N/A	N/A	N/A	N/A	N/A	N/A	-186	+92	N/A	

Nomenclature: See separate page